

2D LIDAR MAPPING

LiStars

Presented by

Zac Swisher, Sneha Patel, Jamil Velez, Bryton Montgomery

SPRINT 1: GOAL

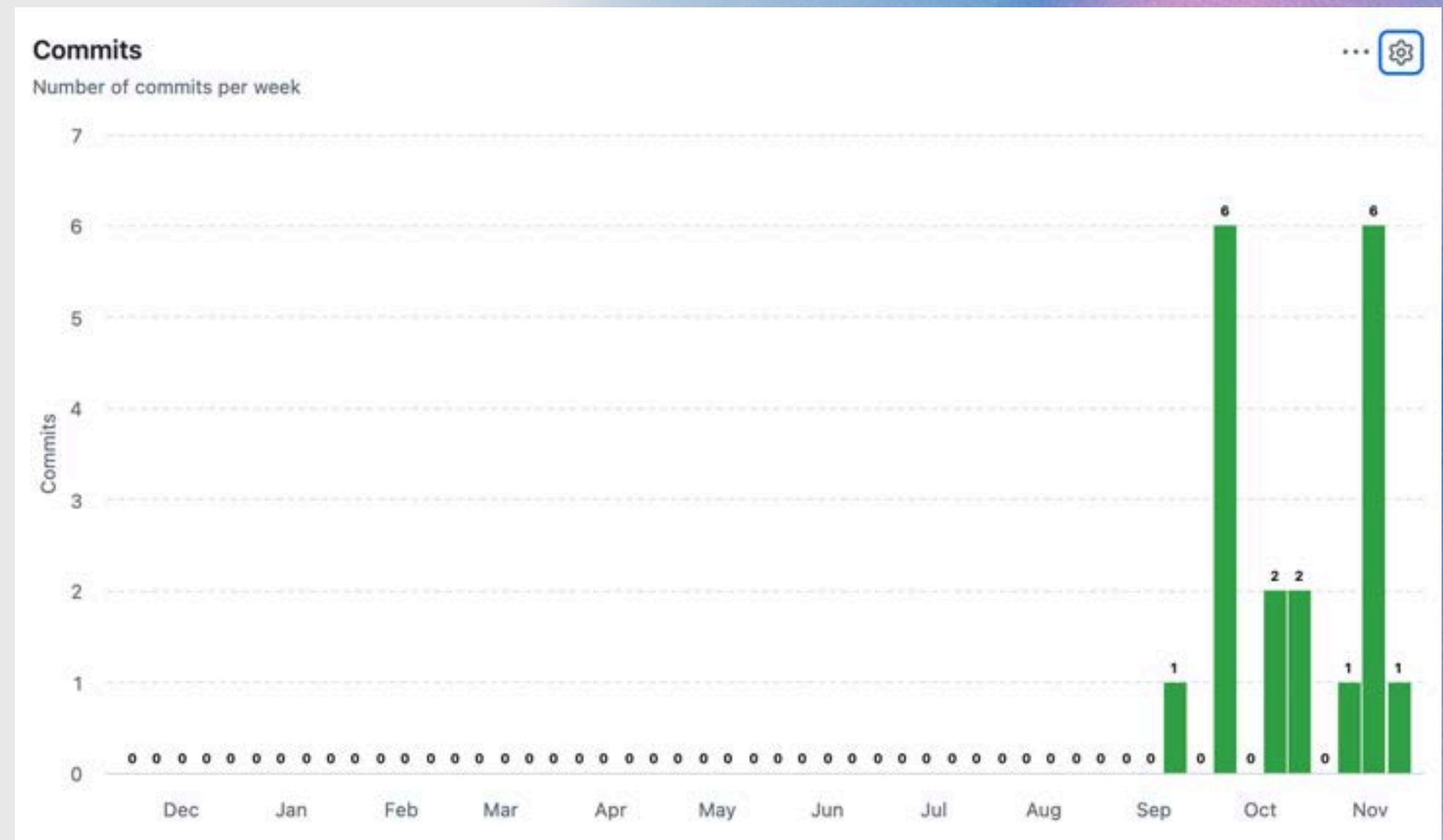
Enable the LiDAR sensor to scan the environment and display detected obstacles in a basic visualization.

Selected Tasks:

- Must Have:
 - Map obstacles via LiDAR sensor
 - Visualize Obstacles
- Should Have:
 - Create Mount for Sensor
- Could have:
 - Assemble LiDAR Sensor & motor on 3D-printed mount

OVERVIEW

- Stand-ups were in person and Discord
- Lessons Learned: Don't over commit, stay on track, documentation, and priorities.
- Adjustments: Time management, clarify task management, and use sprint retrospectives to evaluate progress



Commits over time

Weekly from Sep 6, 2025 to Nov 8, 2025



JamilVelez

11 commits 577 ++ 82 --

#1



swisherrr

5 commits 379 ++ 272 --

#2



sneha0832

2 commits 13 ++ 13 --

#3



Ghostb9

1 commit 2 ++ 2 --

#4



This board is set to public. You can change its visibility at any time. [More about public boards](#)

Sprint 1

BMJVJGSPZ

Story Points

Reports-Beta

Card Dependencies

Product Backlog

As a user, I want to be able to map obstacles in an area.

As a user, I want to be able to view obstacles in a convenient GUI

+ Add a card

Sprint Backlog

To-Do

Sprint Goal: Enable the LiDAR sensor to scan the environment and display detected obstacles in a basic visualisation

+ Add a card

In progress

In progress

+ Add a card

Testing/Review

+ Add a card

Done

Done

README.md

Requirements

Map obstacles via LiDAR sensor

Visualize Obstacles

Create Mount for Sensor

Assemble LiDAR Sensor & motor on 3D-printed mount

+ Add a card

+ Add another list

Inbox

Planner

Board

Switch boards

Task 1: Map Obstacles Via LiDAR Sensor



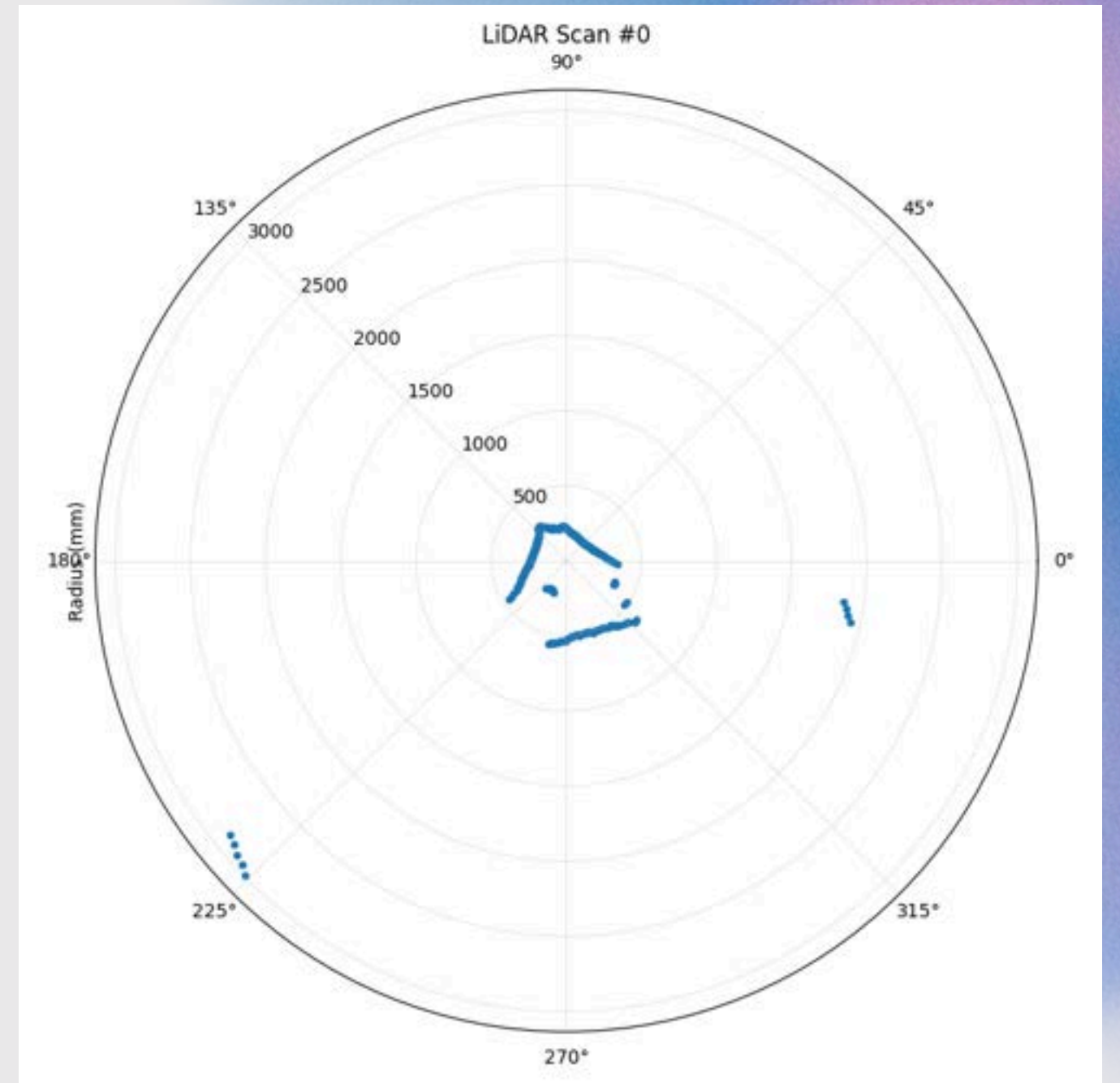
timestamp	scan_idx	meas_idx	angle_deg	distance_mn	quality	x_mm	y_mm
2025-11-02T	0	0	0.562	222.5	15	222.5	2.2
2025-11-02T	0	1	0.578	220.8	15	220.7	2.2
2025-11-02T	0	2	3.047	224.2	15	223.9	11.9
2025-11-02T	0	3	4.281	226.2	15	225.6	16.9
2025-11-02T	0	4	5.516	228.8	15	227.7	22
2025-11-02T	0	5	6.234	232.5	15	231.1	25.2
2025-11-02T	0	6	6.5	230.5	15	229	26.1
2025-11-02T	0	7	9.219	235.2	15	232.2	37.7
2025-11-02T	0	8	9.953	238	15	234.4	41.1
2025-11-02T	0	9	10.406	240.5	15	236.5	43.4
2025-11-02T	0	10	12.922	243.2	15	237.1	54.4
2025-11-02T	0	11	14.156	246.5	15	239	60.3
2025-11-02T	0	12	14.391	250	15	242.2	62.1
2025-11-02T	0	13	14.938	253.8	15	245.2	65.4
2025-11-02T	0	14	15.594	257.5	15	248	69.2
2025-11-02T	0	15	17.516	261.5	15	249.4	78.7
2025-11-02T	0	16	18.125	265.2	15	252.1	82.5
2025-11-02T	0	17	20.234	269.5	15	252.9	93.2
2025-11-02T	0	18	20.891	274	15	256	97.7
2025-11-02T	0	19	21.641	279	15	259.3	102.9
2025-11-02T	0	20	23.234	284	15	261	112
2025-11-02T	0	21	24.328	289.8	15	264	119.4
2025-11-02T	0	22	24.75	296	15	268.8	123.9
2025-11-02T	0	23	25.922	301.8	15	271.4	131.9
2025-11-02T	0	24	27.25	308.5	15	274.3	141.3
2025-11-02T	0	25	28.812	314.8	15	275.8	151.7
2025-11-02T	0	26	29.906	322.2	15	279.3	160.7
2025-11-02T	0	27	30.906	329.5	15	282.7	169.2

Task 1: Map Obstacles Via LiDAR Sensor

Effort Estimation: 7 Story Points

Actual: 7 Story Points

Task 2: Visualize Obstacles

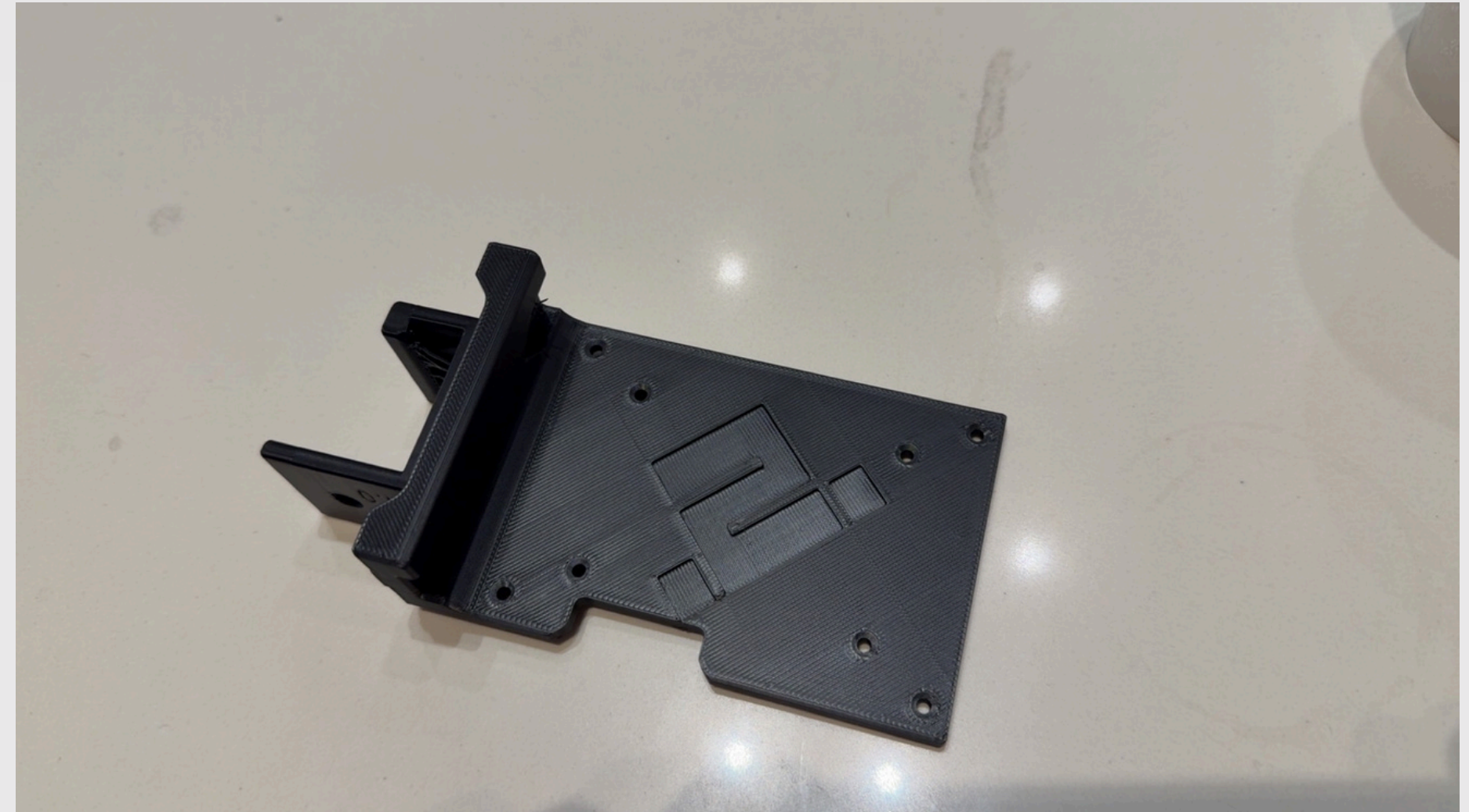
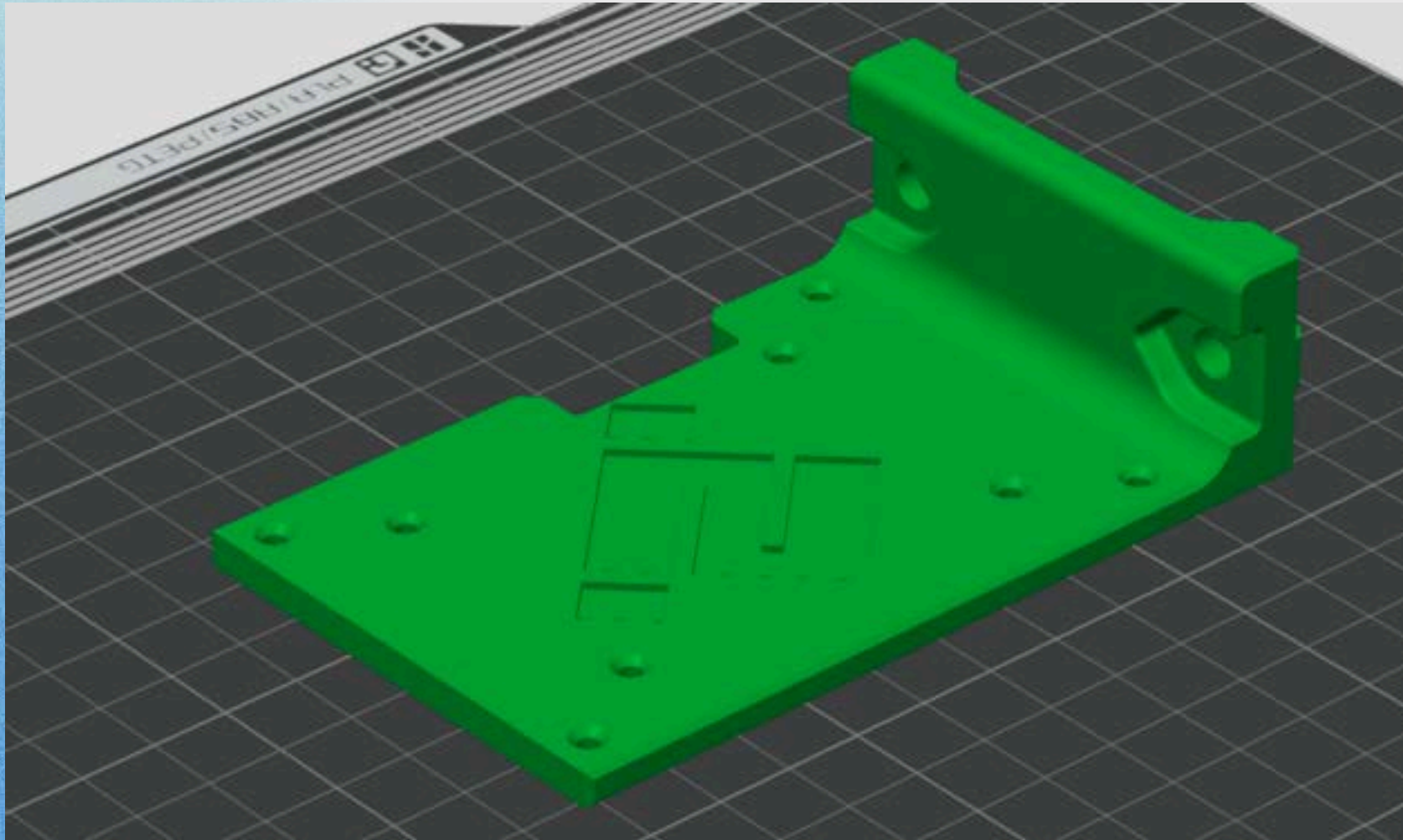


Task 2: Visualize Obstacles

Effort Estimation: 2 Story Points

Actual: 2 Story Points

Task 3: Create Mount for Sensor



Task 3: Create Mount for Sensor

Effort Estimation: 2 Story Points

Actual: 4 Story Points

Task 4: Assemble LiDAR Sensor & motor on 3D-printed mount



Task 4: Assemble LiDAR Sensor & motor on 3D-printed mount

Effort Estimation: 1 Story Point

Actual: 1 Story Point

LIVE DEMO

SPRINT 2

Sprint Goal: Create a functional prototype that can scan obstacles while moving, is housed in a basic enclosure, and can export the collected data.

Selected Tasks:

- Must Have:
 - Assemble LiDAR Sensor & motor on 3D-printed mount
- Should Have:
 - Maps obstacles within radius 360 degrees

Outcome: By the end of Sprint 2, the system should physically rotate and generate accurate 360° obstacle maps.